

[Skip to main content](#)
[Enable accessibility for low vision](#)
[Open the accessibility menu](#)



CodePath Courses

Courses

Quick Links

Quick Actions



Giancarlo Montes

CYB102 | Intermediate Cybersecurity

Intermediate Cybersecurity Fall 2024 (@ Section 1 | Tuesdays 5PM - 7PM PDT)

Personal Member ID#: **113724**

Need help? Post on our [class slack channel](#) or email us at support@codepath.org

- Getting Started
- Course Overview
- Schedule
- VM Access Guide
- AI Usage
- My Report
- Individual Projects
 - Unit 1
 - Unit 2
 - Unit 3
 - Unit 4
 - Unit 5
 - Unit 6

Group Capstone

Groupings
Career Resources

Pework

Overview
Lab
Project
Resources
Career

Lab #7: Splunking for Threats

🚩 MACHINE ACCESS CHECK 🚩

To proceed with this unit, please ensure you have access to a **cyb102 Ubuntu VM** as detailed in the [🔗 VM Access Guide](#).

We recommend you begin with **Azure Labs**.

🛑 This unit is **incompatible with m-chip Mac / ARM architecture**. If your computer has this processor, you will need to choose a **cloud-based VM option** in the [🔗 VM Access Guide](#).

Overview

In this lab, we'll be revisiting our SIEM tool, [Splunk](#). If you remember SIEM (or *Security Information and Event Management*) systems allow us to collect, analyze, and visualize data from various sources to help us identify potential security incidents.

For this unit, we're concerned with **information about threats**. But, conveniently for us, that information is also **data**, which means we can use our SIEM to handle it.

First, we'll import new threat data into our SIEM. Next, we'll explore this data, learning more about the threats and what they might be doing to our organization. Splunk can be a powerful ally in identifying and combating Advanced Persistent Threats (APTs) and other cyber enemies.

If your **Splunk Enterprise demo license** is expiring, you can switch the demo license to a free license by running:

```
sudo /opt/splunk/bin/splunk edit licenser-groups -name "Free" -is_active "1"  
sudo /opt/splunk/bin/splunk restart
```

in the terminal on your VM. In case you are prompted for credentials, these would be the codepath/codepath credentials used to login to Splunk.

Goals

By the end of this lab, you will be able to:

- Grasp how to feed Splunk with external intelligence sources.
- Navigate through Splunk's analytical prowess to sift out potential threats.
- Create and customize Splunk dashboards for threat monitoring.
- Correlate events within Splunk to identify potential threats.

Resources

- [Splunk Documentation](#)
- [Splunk Reference Guide](#)
- [Free Splunk Fundamentals eLearning Courses](#)
- Lab Files:
 - [Threat Feed \(TorList.csv\)](#)
 - [Sample Log 1 \(NetworkProxyLog01.csv\)](#)

Lab Instructions

Required Features

Step 0: Get back into your Splunk environment

In this step, you'll RDP into your Ubuntu Machine and access the preconfigured Splunk instance through your browser at <http://127.0.0.1:8000>.

1. RDP (not SSH) into the Ubuntu Box using the provided credentials
2. Open a browser and navigate to <http://127.0.0.1:8000>
3. Log in to Splunk using the username `codepath` and password `codepath`

 **Checkpoint 0:** You should now be logged in to Splunk and able to view the main dashboard.

Step 1: Let's feed ourselves

Q: What is a Threat Intelligence Feed and why are they necessary?

Q: What types of feeds are out there?

We're going to bring these into Splunk so we can use these feeds to expand our searching capabilities.

For our purposes, we'll be working with a **static threat intelligence feed**.

 **Test Yourself:** What do you think "**static**" means in this context?

1.1. Initiate the Upload

Begin by navigating to `Settings > Add Data` in your Splunk instance. This is the starting point where Splunk allows you to introduce new data for analysis.

If you like, think of it as introducing a new folder of evidence into your favorite detective story – this is the information your Splunk environment will investigate.

1.2. Select Your Data

Click on [Upload files](#) from the options provided. This step is where you tell Splunk, Browse to where your [threat intelligence CSV file](#) is located and select it for upload.

- Click for Raw URL

1.3. Configure the Data Settings

After uploading the file, Splunk will ask you to set the source type. Since we're dealing with [CSV files](#), you can either choose "csv" from the list or let Splunk auto-detect it. This tells Splunk how to interpret the data format, ensuring that it correctly parses the fields and records.

1.4. Input Settings

When you get to Input Settings we can change the field that that host shows up for. This normally lets us tell what computer this data comes from. In this case let's call it Threat Feed so we don't mix it up with our normal data sources.

Add Data

Select Source

Set Source Type

Input Settings

Review

Done

< Back

Review >

Input Settings

Optionally set additional input parameters for this data input as follows:

Host

When the Splunk platform indexes data, each event receives a "host" value. The host value should be the name of the machine from which the event originates. The type of input you choose determines the available configuration options. [Learn More](#)

☒ Constant value

☐ Regular expression on path

☐ Segment in path

Host field value

Threat Feed

Index

The Splunk platform stores incoming data as events in the selected index. Consider using a "sandbox" index as a destination if you have problems determining a source type for your data. A sandbox index lets you troubleshoot your configuration without impacting production indexes. You can always change this setting later. [Learn More](#)

Index

Default

Create a new index

FAQ

> How do indexes work?

> How do I know when to create or use multiple indexes?

1.5. Review and Complete

Before finalizing the upload, you'll have the opportunity to review the data and make sure everything looks correct. Once you've finished uploading the file you'll be able to start searching.

- (Tip: Notice the "Splunk Search" that was made for this data.)

 **Checkpoint 1:** Congratulations detective, you've prepared your evidence for analysis! Go to the Search & Reporting app and try running a basic search, like `source="TorList.csv"` to view the ingested data.

This is where the real investigation begins – you'll start to uncover patterns, identify potential threats, and understand how this static data can be used to simulate real-world cybersecurity scenarios.

Step 2: Analyzing Threat Intelligence Data

With your data now in Splunk, it's time to turn this information into insight. You're the proverbial detective, combing through clues to find patterns that reveal a criminal's plan.

2.1. Start Exploring

Start exploring the data we made in Step 1. Use what you know about threats to make your best guess at where to look -- right now we're just exploring!

To start with, try looking at things that identify the actor and the file: IPs, URLs, file hashes. What fields are parsed out?

Q: Okay, but what even IS this threat feed we uploaded? (Click to expand)



Test Yourself: *Why do you think we're adding this feed?*

-
-
-

Now that we have a threat feed in Splunk this is data we can use over and over. Some threat feeds are static information, some threat feeds are updated dynamically, and others are Indicators of Compromise (IOCs) that we use to investigate new and upcoming threats. And for even more power, we can combine **multiple sources of information**.

2.2. New Data

In 2.1, a Q&A box explains the first file we imported: a list of IP addresses associated with the TOR network. Now, let's bring in some (fictional) network user data to see if any of our users were using TOR... which could indicate suspicious activity on our network.

Go ahead and import [NetworkProxyLog01.csv](#) into Splunk.

- (Tip: You can use all the default values!)
- Click for Raw URL

In our Threat Feeds, all the IPs were imported as `IP Address`. Fortunately, our Network proxy data also uses this field! This will make it easy to search the data for these values.

2.3. By Their Powers Combined

For our next step, we need to build a search that searches **BOTH** of these sources we've imported and see if any of the IP addresses show up **more than once** in **both sources**.

 **Test Yourself:** *Try writing the search command yourself! Then click this text to view our version.*

Our Splunk search command does a few critical things:

1. It looks for entries in both the threat feed (`TorList.csv`) and the network proxy log (`NetworkProxyLog01.csv`).
2. It aggregates the data by the "IP Address" field, which is common between the threat feed and the proxy log.
3. It filters the results to only show IP addresses that appear more than once across both data sources. In the context of our investigation, this means an IP address from the TOR network (a potential threat) has been active in our network, as recorded by the proxy log.

 **Test Yourself:** *What does it mean if we find a match?*

Let's click on count and figure out who was using TOR in our network.

Notice we have fields that are useful to us such as the Computer Name, User Agent String, Date & Time.

 **Checkpoint 2:** At this point, you should have uncovered your prime suspect -- a user who seems to be using the TOR network. You've also gathered some additional information about them, such as the computer they're using and the date and time of their activity.

This is a critical step in the investigation, as it gives you a starting point for understanding how these threat indicators interact with your network.

Stuck? Click [here](#) to view a screenshot of the user fields we found.

Type	☒ Field	Value	Action
Selected	☒ host ▾	WebProxy01	▾
	☒ source ▾	NetworkProxyLog01.csv	▾
	☒ sourcetype ▾	csv	▾
Event	☐ Action ▾	Allowed	▾
	☐ Bytes Received ▾	54035	▾
	☐ Bytes Sent ▾	54320	▾
	☐ Computer Name ▾	Workstation-008	▾
	☐ Date ▾	10/19/2024	▾
	☐ IP Address ▾	140.78.100.14	▾
	☐ Source IP ▾	10.10.10.162	▾
	☐ Time ▾	15:13:37	▾
	☐ URL ▾	https://example.com/	▾
	☐ User Agent String ▾	Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/605.1.15 (KHTML, like Gecko) Version/14.0.3 Safari/605.1.15	▾
Time ⚙	☐ _time ▾	2024-03-04T15:13:37.100+00:00	
Default	☐ index ▾	main	▾

Step 3: Visualizing the Data with a Dashboard

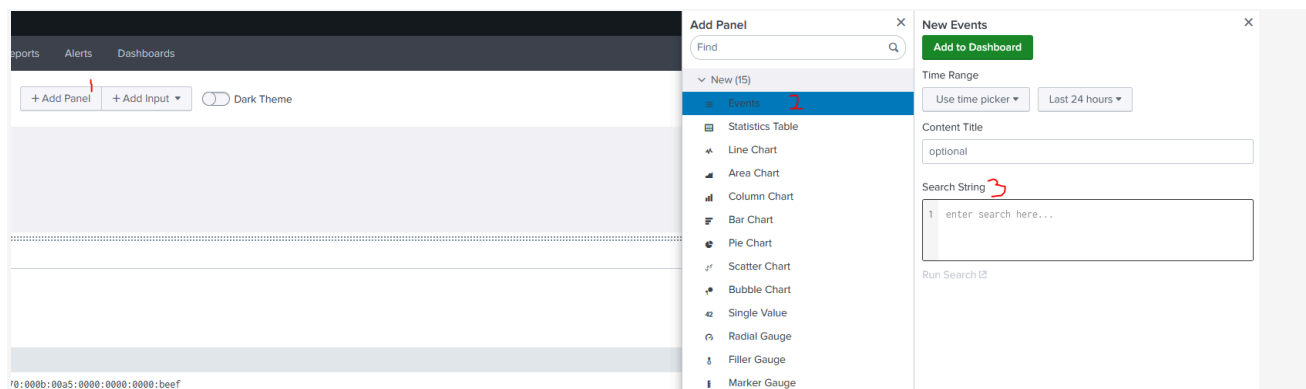
Once you understand how these threat indicators interact with your network, creating a dashboard can help you visualize and monitor these interactions over time instead of searching each and every time. A shortcut to your investigation!

3.1. Creating a New Dashboard

Navigate to the Dashboards section in Splunk and create a new dashboard titled "Threat Intelligence Monitoring".

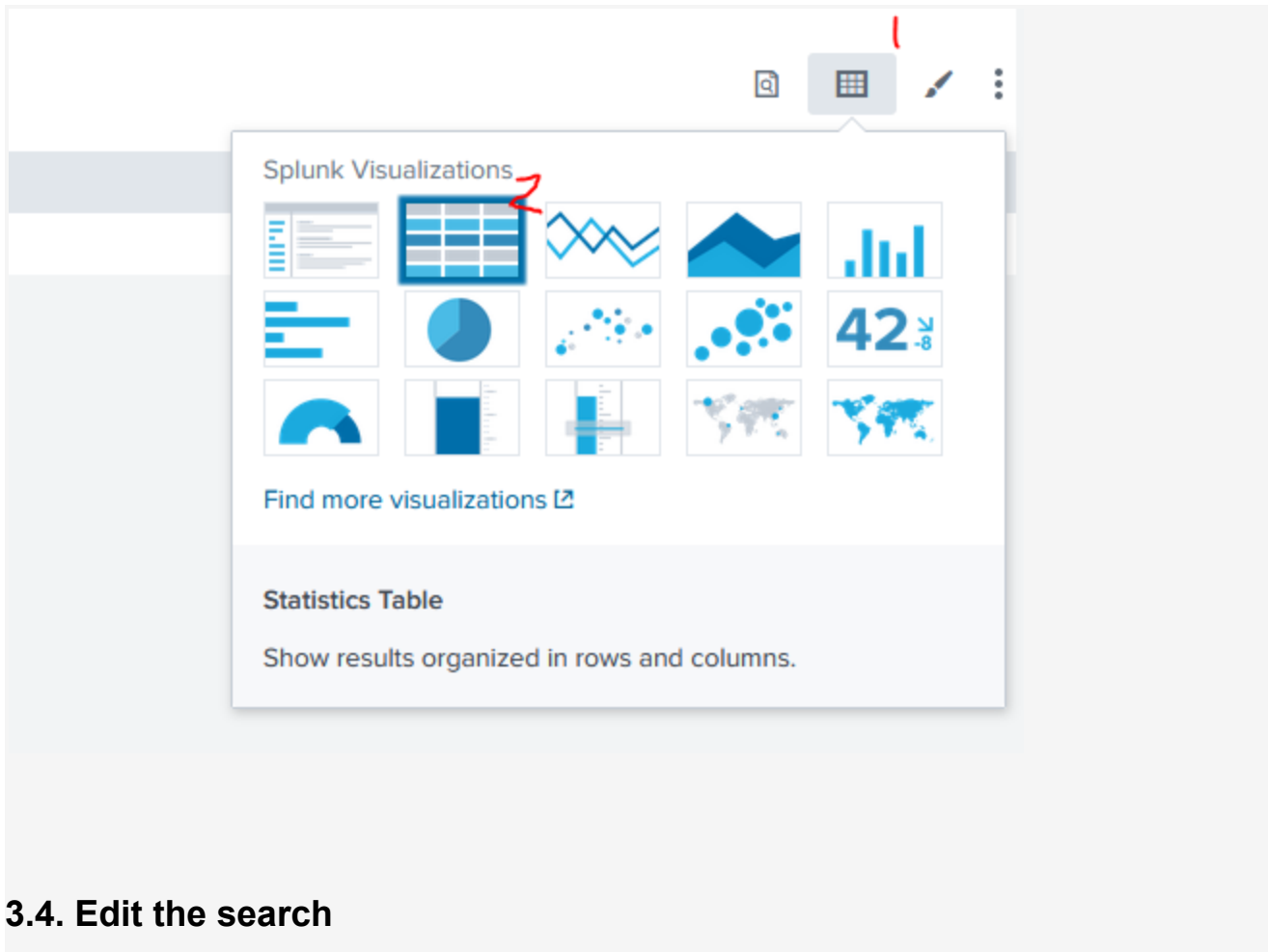
3.2. Adding a Panel

Add a panel that uses the search query you just created by clicking Add Panel > New > Events. Then paste in the search from above. This panel will display all instances where known TOR network IP addresses have accessed your network.



3.3. Adjust the visualization

You might notice that the visualization is pulling all events. If you click on the icon next to the magnifying glass (Select Visualization) you can change the way the data shows up. We're going to change it to "Statistics Table"



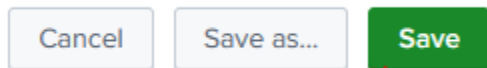
3.4. Edit the search

Click the Magnifying Glass to edit the search. Set the Auto Refresh Delay to 1 minute & make sure the Time Range is Other> All time so we don't miss the logs.
Replace your search with the following search:

```
(index=main source="TorList.csv") OR (index=main  
source="NetworkProxyLog01.csv")  
| stats values(source) as sources, values("Computer Name") as ComputerName,  
values("User Agent String") as UserAgent, values(Date) as Date, values(Time) as  
Time by "IP Address"  
| where mvcount(sources) > 1  
| table "IP Address", ComputerName, UserAgent, Date, Time
```

Note: This Splunk search uses some advanced options to adjust the fields that are beyond the scope of this course, you can learn more about this by checking out the [Splunk documentation](#).

3.5. Let's save this: Now our dashboard is much friendlier, let's click the save button in the top right.



@ Checkpoint 3: Now we've got a Dashboard that shows whenever something matches data from our threat feed from the last 24 hours. In a production environment you'd have user access logs constantly feeding into Splunk. So often, a SOC will use threat feeds to monitor for suspicious behaviors.

With a dashboard, you could make panels for ALL your various threat feeds and have your SOC investigate the results as they come in. This is how an organization can respond to things going on mere *moments* after they happen.

Step 4: Creating a Report from Your Dashboard

While dashboards are excellent for real-time monitoring, reports are crucial for communicating your findings to others in your organization.

4.1. Generating a Report:

Let's continue from our Dashboard and click edit to review the panels.

Click the magnifying glass to edit your search and select "Convert to Report"

Edit Search

Title

Search String

```
1 (source="TorList.csv" ) OR source="networkproxylog01.csv" )
2 | stats values(source) as sources, values("Computer Name") as ComputerName,
  values("User Agent String") as UserAgent, values(Date) as Date, values
  (Time) as Time by "IP Address"
3 | where mvcount(sources) > 1
4 | table "IP Address", ComputerName, UserAgent, Date, Time
```

Run Search [?](#)

Time Range

Use time picker ▼

Last 24 hours ▶

Auto Refresh Delay ?

1 minute ▼

Refresh Indicator

Progress bar ▼

Cancel

Convert to Report

Apply

4.2 Name your report:

Give it a useful name! We chose "Tor Search". When you're done, click "Apply".
Click the reports tab to view your report!

Tip: In Enterprise Environments you'd have this report automatically emailed out, or even potentially automatically interact with your tools to act upon the threat feeds.

As the digital landscape adapts threat feeds will adapt and change. Some will provide Indicators of Compromise, some will just provide you things to be aware of, and others may look for threats specific to your organization. Keep your eyes open for new feeds and new ways to use them, detective, and don't be afraid to get creative!



You've completed your lab! You now know the basics of using Splunk to monitor for threats!

Stretch Features

Want to go further? Here are some ideas!

- Do an online search in the search engine of your choosing and look for free Threat Feeds. Try to import one into Splunk and see if you can match any interesting data.
- Create new dashboards and reports for other threat feeds. You can use the same search command and just change the source.
-